The EFGP-SING E-step

GP regression on uncertain latent velocities, solved by Fourier features

Abstract

This note derives the E-step of SING with a GP drift prior, accelerated by EFGP. The derivation is laid out in four strict layers, introduced once each: **(1)** the conceptual update is GP regression on latent velocities; **(2)** when the latent path is uncertain, fixed sufficient statistics are replaced by expectations under $q(x)$, with the linear term resolved by Stein's lemma; **(3)** EFGP is the kernel approximation $K(x, x') \approx \phi(x)^* \phi(x')$, which turns the same regression problem into a matrix-free system $A_r \mu_r = h_r$; **(4)** the expectations are computed by Monte Carlo over $q(x)$ using NUFFT and BTTB-Toeplitz primitives.

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1 Setup

Latent SDE and observations

We assume an Euler-discretised latent SDE

$$x_{i+1} \;=\; x_i \;+\; \Delta_i\, f(x_i) \;+\; \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0,\, \Delta_i \Sigma), \tag{1}$$

with Gaussian observations $p(y_i \mid x_i)$. Throughout, we abbreviate

$$\delta_i \;:=\; x_{i+1} - x_i. \tag{2}$$

The diffusion is taken diagonal, $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_{D_x}^2)$. The drift coordinates carry independent GP priors,

$$f_r \sim \mathrm{GP}(0, k_\theta), \qquad r = 1, \ldots, D_x. \tag{3}$$

Joint and ELBO

The joint density factorises as

$$p(f, x_{0:T}, y_{0:T}) \;=\; p(f)\, p(x_0) \prod_i p(x_{i+1} \mid x_i, f) \prod_i p(y_i \mid x_i), \tag{4}$$

and SING uses a structured variational posterior $q(x_{0:T})\, q(f)$. The ELBO splits into

$$\mathcal{L}(q) \;=\; \underbrace{\mathbb{E}_{\log p(y \mid x)}}_{\text{depends only on } q(x)} \;+\; \underbrace{\mathbb{E}_{\log p(x \mid f)}}_{\text{couples } q(x),\, q(f)} \;+\; \underbrace{\mathbb{E}_{\log p(f)} - \mathbb{E}_{\log q(f)}}_{\text{prior + entropy of } q(f)} + (\text{terms in } q(x) \text{ only}). \tag{5}$$

Only the transition term $\mathbb{E}_{\log p(x \mid f)}$ couples $q(x)$ and $q(f)$. Everything in this document follows from analysing that one term.

2 The transition term, stripped to its f-content

The Euler transition log-density is

$$\log p(x_{i+1} \mid x_i, f) \;=\; -\frac{1}{2\Delta_i} \big[\delta_i - \Delta_i\, f(x_i)\big]^\top \Sigma^{-1} \big[\delta_i - \Delta_i\, f(x_i)\big] \;+\; \text{const.} \tag{6}$$

The constant and the $\delta_i^\top \Sigma^{-1} \delta_i$ term do not involve f. Expanding for diagonal Σ, the part of $\log p(x_{i+1} \mid x_i, f)$ that depends on coordinate f_r is

$$\boxed{\frac{1}{\sigma_r^2}\, \delta_{i,r}\, f_r(x_i) \;-\; \frac{\Delta_i}{2\sigma_r^2}\, f_r(x_i)^2.} \tag{7}$$

This is a Gaussian likelihood, quadratic in the function values $f_r(x_i)$. That is the *only* fact about the transition we will need. The $q(f_r)$ update reduces to combining (7) with the GP prior (3).

Independence across r. Because Σ is diagonal and the priors on f_r are independent, the $q(f_r)$ updates decouple across r. From now on we work coordinate-wise and suppress the index r when convenient.

3 Layer 1: GP regression with known x

Pretend, for this section only, that the latent path $X = \{x_0, \ldots, x_{T-1}\}$ is observed. Define the pseudo-velocity

$$y_{i,r}^{\text{pseudo}} \;:=\; \frac{\delta_{i,r}}{\Delta_i} \;=\; \frac{x_{i+1,r} - x_{i,r}}{\Delta_i}. \tag{8}$$

Completing the square in (7) shows the contribution from transition i is

$$-\frac{\Delta_i}{2\sigma_r^2}\big(y_{i,r}^{\text{pseudo}} - f_r(x_i)\big)^2 \;+\; \text{const}, \tag{9}$$

i.e. a Gaussian observation of $f_r(x_i)$ with precision

$$\lambda_{i,r} \;=\; \frac{\Delta_i}{\sigma_r^2}. \tag{10}$$

Stack the latent function values at the path locations,

$$f_{r,X} \;=\; \big[f_r(x_0), \ldots, f_r(x_{T-1})\big]^\top \in \mathbb{R}^T, \qquad [K_X]_{ij} = k_\theta(x_i, x_j), \qquad \Lambda_r = \operatorname{diag}\big(\lambda_{i,r}\big), \tag{11}$$

and let $y_r^{\text{pseudo}} \in \mathbb{R}^T$ collect the pseudo-velocities. The negative log posterior at $f_{r,X}$ is then

$$\tfrac{1}{2}\, f_{r,X}^\top K_X^{-1} f_{r,X} \;+\; \tfrac{1}{2}\,(y_r^{\text{pseudo}} - f_{r,X})^\top \Lambda_r (y_r^{\text{pseudo}} - f_{r,X}) \;+\; \text{const}, \tag{12}$$

so $q(f_{r,X}) = \mathcal{N}(m_{r,X},\, C_{r,X})$ with the standard GP-regression identities

$$\boxed{C_{r,X}^{-1} \;=\; K_X^{-1} + \Lambda_r, \qquad C_{r,X}^{-1}\, m_{r,X} \;=\; \Lambda_r\, y_r^{\text{pseudo}}.} \tag{13}$$

Takeaway. *The $q(f)$ update is GP regression on latent velocities.* Sections 4–5 will modify (13), but the derivation never escapes this template.

4 Layer 2: uncertain x, in a finite basis

In SING, X is not observed; only $q(x_{0:T})$ is available, a Gaussian Markov variational posterior with marginals

$$m_i = \mathbb{E}[x_i], \qquad S_i = \operatorname{Cov}(x_i), \qquad S_{i,i+1} = \operatorname{Cov}(x_i, x_{i+1}). \tag{14}$$

The K-space packaging (13) no longer applies directly: with X random, both $f_{r,X}$ and K_X are themselves random objects. But the mean-field update for $q(f_r)$ is still well-defined:

$$\log q^*(f_r) \;=\; \log p_\theta(f_r) \;+\; \sum_i \mathbb{E}_{q(x_i,\, x_{i+1})}\left[\frac{\delta_{i,r}\, f_r(x_i)}{\sigma_r^2} \;-\; \frac{\Delta_i\, f_r(x_i)^2}{2\sigma_r^2}\right] \;+\; \text{const}. \tag{15}$$

The expectation here is over x only — f_r is the variable we are updating, so it stays inside.

Pick a basis. Expand f_r in any finite basis,

$$f_r(x) \;=\; \phi(x)^*\, w_r, \qquad w_r \sim \mathcal{N}(0, I_M), \tag{16}$$

chosen so that the GP prior on f_r becomes the iid prior on the coefficients w_r. (Section 5 specialises ϕ to Fourier features; for now any basis works.) Substituting (16) into (15),

$$\log q^*(w_r) \;=\; -\tfrac{1}{2}\, w_r^* A_r\, w_r \;+\; \Re\big(h_r^* w_r\big) \;+\; \text{const}, \tag{17}$$

with

$$\boxed{A_r \;=\; I + \frac{1}{\sigma_r^2}\sum_i \Delta_i\, \mathbb{E}_{q(x_i)}\big[\phi(x_i)\phi(x_i)^*\big], \qquad h_r \;=\; \frac{1}{\sigma_r^2}\sum_i \mathbb{E}_{q(x_i,\, x_{i+1})}\big[\phi(x_i)\,(x_{i+1,r} - x_{i,r})\big].} \tag{18}$$

The integration over x contracts $\phi\phi^*$ and ϕ down to the deterministic $M \times M$ matrix and M-vector inside (18); the variable w_r is never touched. So $\log q^*(w_r)$ is manifestly quadratic in w_r, and

$$q(w_r) \;=\; \mathcal{N}(\mu_r,\, A_r^{-1}), \qquad A_r\,\mu_r \;=\; h_r. \tag{19}$$

This is the EFGP analogue of the K-space identities (13), under the substitutions $K_X^{-1} + \Lambda_r \leftrightarrow A_r$ and $\Lambda_r y_r^{\text{pseudo}} \leftrightarrow h_r$. The expectation over $q(x)$ neither destroys w_r nor breaks Gaussianity; it only determines the deterministic coefficients of the system.

Stein identity for the linear term in h_r

The linear coefficient in (18) is a pairwise expectation under $q(x_i, x_{i+1})$, which is awkward. Stein's lemma collapses it to marginal expectations plus a covariance correction. Let

$$d_i \;:=\; m_{i+1} - m_i, \qquad C_i \;:=\; S_{i,i+1} - S_i \;=\; \mathrm{Cov}(x_i,\, x_{i+1} - x_i). \tag{20}$$

For any sufficiently smooth g (vector- or scalar-valued), the Gaussian Stein identity gives

$$\mathbb{E}\big[g(x_i)\,(x_{i+1,r} - x_{i,r})\big] \;=\; \mathbb{E}[g(x_i)]\,d_{i,r} \;+\; \mathbb{E}[J_g(x_i)]\,C_{i,\cdot r}, \tag{21}$$

where $C_{i,\cdot r}$ is the r-th column of C_i. Applying this to $g = \phi$ (componentwise) inside (18),

$$\boxed{h_r \;=\; \frac{1}{\sigma_r^2}\sum_i \Big\{\underbrace{\mathbb{E}_{q(x_i)}[\phi(x_i)]\,d_{i,r}}_{\text{mean-increment}} \;+\; \underbrace{\mathbb{E}_{q(x_i)}[J_\phi(x_i)]^*\,C_{i,\cdot r}}_{\text{Stein correction}}\Big\}.} \tag{22}$$

After this collapse, every quantity in (19) depends only on *marginal* expectations under $q(x_i)$, plus the cross-cov correction C_i.

5 Layer 3: EFGP — the Fourier-feature choice

EFGP specialises the basis ϕ in (16) to Fourier features:

$$\phi_\theta(x) \;=\; D_\theta\,F_x, \qquad F_x[k] \;=\; e^{2\pi i\, x^\top \xi_k}, \qquad \phi_\theta(x)^*\phi_\theta(x') \;\approx\; k_\theta(x, x'), \tag{23}$$

where $\{\xi_k\}_{k=1}^M$ is an equispaced spectral grid and D_θ is diagonal with square-root spectral weights chosen to match k_θ. This is the only kernel approximation introduced. The system $A_r\,\mu_r = h_r$ from (19) is unchanged; what we gain are two structural simplifications.

Cheap Jacobian. Differentiating (23),

$$\frac{\partial \phi_k(x)}{\partial x_j} \;=\; 2\pi i\,\xi_{k,j}\,\phi_k(x), \tag{24}$$

so J_ϕ is just frequency-multiplication of ϕ. The Stein correction in (22) costs no more than evaluating ϕ itself.

BTTB structure of $\mathbb{E}_{q(x_i)}[\phi\phi^*]$. For Fourier features, $\phi(x)\phi(x)^*$ has (k,ℓ) entry $D_\theta\, e^{2\pi i x^\top(\xi_\ell-\xi_k)}\, D_\theta$. The expectation under $q(x_i)$ depends on $\xi_\ell - \xi_k$ only:

$$\big(\mathbb{E}_{q(x_i)}[\phi\phi^*]\big)_{k\ell} \;=\; D_{\theta,k}\, \mathbb{E}_{q(x_i)}\big[e^{2\pi i\, x_i^\top(\xi_\ell-\xi_k)}\big]\, D_{\theta,\ell}. \tag{25}$$

On an equispaced grid this makes $\mathbb{E}_{q(x_i)}[\phi\phi^*]$ Block-Toeplitz with Toeplitz Blocks (BTTB), so $A_r v$ costs $O(M\log M)$ via FFT instead of $O(M^2)$.

Where we are. The variational objective (15) and the resulting linear system (19)–(22) are basis-agnostic. EFGP is the choice of basis that makes both pieces of A_r and h_r computable as Fourier transforms of weighted point clouds — the topic of the next section.

6 Layer 4: computing the expectations

The remaining ingredients are the marginal expectations $\mathbb{E}_{q(x_i)}[\phi(x_i)\phi(x_i)^*]$, $\mathbb{E}_{q(x_i)}[\phi(x_i)]$, and $\mathbb{E}_{q(x_i)}[J_\phi(x_i)]$. These are evaluated by Monte Carlo over $q(x_i)$, then assembled via NUFFT and BTTB-Toeplitz primitives.

Monte Carlo cloud

For each $i = 0,\dots,T-2$ draw S marginal samples

$$x_i^{(s)} \sim q(x_i), \qquad s = 1,\dots,S, \tag{26}$$

and stack them as $X = \{x_i^{(s)}\}_{i,s} \in \mathbb{R}^{(T-1)S\times D_x}$. Define

$$[F_X]_{k,(i,s)} \;=\; e^{2\pi i\,(x_i^{(s)})^\top \xi_k}, \qquad \phi_\theta(x_i^{(s)}) \;=\; D_\theta\,(F_X)_{:,(i,s)}. \tag{27}$$

The precision A_r as a Toeplitz operator

Set the per-sample weight

$$(W_r)_{(i,s)} \;=\; \frac{\Delta_i}{S\,\sigma_r^2}. \tag{28}$$

Then A_r in (18) becomes

$$A_r \;\approx\; I + D_\theta\, T_r\, D_\theta, \qquad T_r \;:=\; F_X^*\, W_r\, F_X. \tag{29}$$

Crucially, the (k,ℓ) entry of T_r depends on the frequency *difference* $\xi_\ell - \xi_k$:

$$(T_r)_{k\ell} \;=\; \sum_{i,s} \frac{\Delta_i}{S\sigma_r^2}\, e^{2\pi i\,(x_i^{(s)})^\top(\xi_\ell-\xi_k)}. \tag{30}$$

For an equispaced spectral grid, T_r is therefore Block-Toeplitz with Toeplitz Blocks (BTTB), and a single Type-1 NUFFT over the cloud X produces its generating kernel. Matvecs $A_r v$ are

$$A_r v \;=\; v + D_\theta\big(T_r\,(D_\theta v)\big), \tag{31}$$

implemented in the code as `make_A_apply` (see `sing/efgp_jax_primitives.py`). The system $A_r\mu_r = h_r$ is then solved by Jacobi-preconditioned conjugate gradient.

Building h_r

Define

$$(a_r)_{(i,s)} \;=\; \frac{d_{i,r}}{S\,\sigma_r^2}, \qquad (c_{j,r})_{(i,s)} \;=\; \frac{[C_i]_{j,r}}{S\,\sigma_r^2}. \tag{32}$$

Then (22) estimates as

$$h_{1,r} \;\approx\; D_\theta\, F_X^* a_r, \qquad h_{2,r} \;\approx\; \sum_{j=1}^{D_x} (-2\pi i\,\xi_j) \odot D_\theta\, F_X^* c_{j,r}, \qquad h_r = h_{1,r} + h_{2,r}. \tag{33}$$

Each term costs one Type-1 NUFFT plus a frequency-weighted reduction.

Sign of the Stein correction. The factor in $h_{2,r}$ is $-2\pi i\,\xi_j$ (negative) because the term comes from the *adjoint* Jacobian J_ϕ^*, and $\overline{2\pi i\,\xi_{k,j}\,\phi_k(x)} = -2\pi i\,\xi_{k,j}\,\overline{\phi_k(x)}$. This sign is straightforward to miss from (24) alone — see `compute_mu_r_jax` in `sing/efgp_jax_drift.py` (and the `EFGP_SING_README.md` bug-fix history).

Cross-covariance convention. SING stores $\texttt{SSs}[t] = \mathrm{Cov}(x_{t+1}, x_t)$, but the Stein identity needs $S_{i,i+1} = \mathrm{Cov}(x_i, x_{i+1})$, which is $\texttt{SSs}[t]^\top$.

Pair-sampling remark

The Stein form (22) can be checked against a naive pair-sampling estimator: draw transition pairs $(x_i^{(s)}, x_{i+1}^{(s)}) \sim q(x_i, x_{i+1})$ and use sampled pseudo-velocities $y_{i,r}^{(s)} = (x_{i+1,r}^{(s)} - x_{i,r}^{(s)})/\Delta_i$ in (13). Both estimators target the same h_r; Stein replaces sampled increments by analytic mean/covariance corrections and has lower variance. SING uses Stein in production and pair-sampling only as a regression test.

7 Posterior predictive moments

The $q(x)$ update needs three drift expectations at each x_i: $\bar{f}(x)$, $\mathbb{E}_{q(f)}[f(x)^\top \Sigma^{-1} f(x)]$, and the Jacobian $J_{\bar{f}}(x)$. With $q(w_r) = \mathcal{N}(\mu_r, A_r^{-1})$,

$$\bar{f}_r(x) \;=\; \phi(x)^* \mu_r, \tag{34}$$

$$\frac{\partial \bar{f}_r}{\partial x_j}(x) \;=\; \phi(x)^* \big[(2\pi i\,\xi_j) \odot \mu_r\big], \tag{35}$$

$$\mathrm{Cov}_{q(f)}\big[f_r(x), f_r(x')\big] \;=\; \phi(x)^* A_r^{-1} \phi(x'). \tag{36}$$

For diagonal Σ and independent output GPs,

$$\mathbb{E}_{q(f)}\big[f(x)^\top \Sigma^{-1} f(x)\big] \;=\; \sum_r \sigma_r^{-2} \big[\, \bar{f}_r(x)^2 + \phi(x)^* A_r^{-1} \phi(x) \,\big]. \tag{37}$$

Mean and Jacobian cost one Type-2 NUFFT each (`drift_moments_jax` in `sing/efgp_jax_drift.py`). The variance $\phi(x)^* A_r^{-1} \phi(x)$ is estimated without forming A_r^{-1} via Hutchinson probes

$$\mathrm{diag}\big(\Phi_{\mathrm{eval}} A_r^{-1} \Phi_{\mathrm{eval}}^*\big) \;\approx\; \frac{1}{J} \sum_{j=1}^{J} z_j \odot \Phi_{\mathrm{eval}} A_r^{-1} \Phi_{\mathrm{eval}}^* z_j, \qquad z_j \sim \text{Rademacher}, \tag{38}$$

each requiring one adjoint Fourier apply, one CG solve with A_r, and one forward Fourier apply (`hutchinson_diag` in `sing/efgp_jax_primitives.py`).

8 Feeding back into the $q(x)$ update

With $q(f)$ frozen, the transition contribution to the ELBO is

$$\mathbb{E}_{q(f)}\big[\log p(x_{i+1} \,|\, x_i, f)\big] \;=\; -\frac{\delta_i^\top \Sigma^{-1} \delta_i}{2\Delta_i} + \delta_i^\top \Sigma^{-1} \bar{f}(x_i) - \frac{\Delta_i}{2}\, \bar{f}(x_i)^\top \Sigma^{-1} \bar{f}(x_i) + \text{const}, \tag{39}$$

where $\bar{f}(x) = \phi(x)^* \mu_r$ is the EFGP posterior-mean drift. SING's natural-gradient update needs the value and the (m_i, S_i)-derivatives of (39) integrated against $q(x_i, x_{i+1}) = \mathcal{N}\big((m_i, m_{i+1}),\, (S_i, S_{i,i+1}; S_{i,i+1}^\top, S_{i+1})\big)$. None of those integrals are closed-form because $\bar{f}$ is nonlinear.

The approximation, in one line. Replace $\bar{f}$ by its first-order Taylor expansion at m_i:

$$\boxed{\widetilde{f}(x_i) \;:=\; \bar{f}(m_i) \;+\; J_{\bar{f}}(m_i)\,(x_i - m_i), \qquad J_{\bar{f}}(m_i)_{rj} \;=\; \left.\frac{\partial \bar{f}_r}{\partial x_j}\right|_{x=m_i}.} \tag{40}$$

$\widetilde{f}$ is affine in x_i. *All* expectations and derivatives below are taken *exactly* after substituting $\widetilde{f}$ for $\bar{f}$ in (39); nothing else is approximated. Curvature of $\bar{f}$ (Hessian and higher) is the only thing dropped.

Expectations under (40). With $x_i \sim \mathcal{N}(m_i, S_i)$, $\widetilde{f}(x_i)$ is Gaussian. Direct computation (using $\mathbb{E}[x_i - m_i] = 0$ and $\mathbb{E}[(x_i - m_i)(x_i - m_i)^\top] = S_i$):

$$\mathbb{E}_{q(x_i)}[\widetilde{f}(x_i)] \;=\; \bar{f}(m_i), \tag{41}$$

$$\mathbb{E}_{q(x_i)}[J_{\widetilde{f}}(x_i)] \;=\; J_{\bar{f}}(m_i) \quad \text{(the linearisation pinned } J \text{ at } m_i\text{)}, \tag{42}$$

$$\mathbb{E}_{q(x_i)}\big[\widetilde{f}(x_i)^\top \Sigma^{-1} \widetilde{f}(x_i)\big] \;=\; \bar{f}(m_i)^\top \Sigma^{-1} \bar{f}(m_i) \;+\; \mathrm{tr}\big[J_{\bar{f}}(m_i)^\top \Sigma^{-1} J_{\bar{f}}(m_i)\, S_i\big]. \tag{43}$$

The cross term in (39) integrates against $q(x_i, x_{i+1})$; the same Stein identity used in §4 (with $\widetilde{f}$ in place of g) gives

$$\mathbb{E}_{q(x_i,\, x_{i+1})}\big[\delta_i^\top \Sigma^{-1} \widetilde{f}(x_i)\big] \;=\; d_i^\top \Sigma^{-1} \bar{f}(m_i) \;+\; \mathrm{tr}\big[\Sigma^{-1} J_{\bar{f}}(m_i)\, C_i^\top\big], \tag{44}$$

so the cross-covariance $S_{i,i+1}$ enters only through $C_i = S_{i,i+1} - S_i$.

Derivatives. The expressions above are explicit functions of (m_i, S_i); differentiate them. The chain rule produces $\partial \bar{f}(m_i)/\partial m_i = J_{\bar{f}}(m_i)$ and $\partial J_{\bar{f}}(m_i)/\partial m_i = H_{\bar{f}}(m_i)$; the Hessian terms are then dropped, since carrying them would re-introduce exactly the curvature that (40) already discarded. (This is the only consistency rule needed: $\bar{f}$ enters at most to first order in $x_i - m_i$, hence $J_{\bar{f}}$ is treated as a constant w.r.t. m_i.) Result:

quantity	value	$\partial/\partial m_i$	$\partial/\partial S_i$
$\mathbb{E}_{q(x_i)}[\widetilde{f}]$	$\bar{f}(m_i)$	$J_{\bar{f}}(m_i)$	0
$\mathbb{E}_{q(x_i)}[J_{\widetilde{f}}]$	$J_{\bar{f}}(m_i)$	0 (Hessian dropped)	0
$\mathbb{E}_{q(x_i)}[\widetilde{f}^\top \Sigma^{-1} \widetilde{f}]$	$\bar{f}^\top \Sigma^{-1} \bar{f} + \mathrm{tr}[J^\top \Sigma^{-1} J\, S_i]$	$2\, J_{\bar{f}}(m_i)^\top \Sigma^{-1} \bar{f}(m_i)$	$J_{\bar{f}}(m_i)^\top \Sigma^{-1} J_{\bar{f}}(m_i)$

Each $\partial/\partial m_i$ uses $\partial \bar{f}(m)/\partial m = J_{\bar{f}}(m)$ and treats J as constant w.r.t. m. Each $\partial/\partial S_i$ uses $\partial\, \mathrm{tr}[A\, S]/\partial S = A$.

Implementation note. The code (`ef_with_jac_grad`, `eff_with_grads` in `sing/efgp_jax_drift.py`) computes the row-3 *value* as just $\bar{f}(m_i)^\top \Sigma^{-1} \bar{f}(m_i)$ — the trace correction in (43) is omitted. The row-3 *gradients* match the linearisation in the table, in particular keeping $\partial/\partial S_i = J_{\bar{f}}^\top \Sigma^{-1} J_{\bar{f}}$ (which is exactly $\partial_{S_i} \operatorname{tr}[J^\top \Sigma^{-1} J\, S_i]$, the S_i-derivative of the trace term).

This asymmetry is purely cosmetic for the optimization:

- $q(x)$ updates are *unchanged* — they consume only the gradients, which are identical to the consistent-linearisation gradients. The "shrink S_i when $J^\top \Sigma^{-1} J$ is large" signal that drives the variational variance toward the drift-sharpness scale is preserved.
- $q(f)$ updates are unchanged (they don't use these moments at all; see §5).
- The M-step is unchanged (it uses the EFGP marginal log-likelihood, not these moments).
- Only the *reported ELBO value* is affected: it is biased low by $\frac{\Delta_i}{2} \operatorname{tr}[J_{\bar{f}}^\top \Sigma^{-1} J_{\bar{f}}\, S_i]$ per transition. This matters for convergence diagnostics, not for the fixed point reached.

Were the trace dropped from the S_i-gradient as well (a strictly weaker zero-th-order approximation), $\partial/\partial S_i$ would collapse to 0 and $q(x)$ *would* change — losing the drift-sharpness pull on S_i. The asymmetric drop deliberately keeps the gradient signal while simplifying the value computation.

Posterior-variance contributions $\mathbb{E}_{q(f)}[f^\top \Sigma^{-1} f] - \bar{f}^\top \Sigma^{-1} \bar{f} = \operatorname{tr}[\Sigma^{-1} \phi(x_i)^* A_r^{-1} \phi(x_i)]$ (Hutchinson) follow the same pattern — included in values when needed, not back-propagated through the CG solve.

9 Algorithm summary

Given current $q(x_{0:T})$ (marginals m_i, S_i and cross-covariances $S_{i,i+1}$), kernel hyperparameters θ, diffusion Σ, spectral grid $\{\xi_k\}$ with weights D_θ, sample count S, and Hutchinson probe count J:

1. **Increments.** $d_i = m_{i+1} - m_i, \quad C_i = S_{i,i+1} - S_i$.
2. **Cloud.** Sample $x_i^{(s)} \sim q(x_i)$ for $i = 0, \ldots, T-2$, $s = 1, \ldots, S$.
3. **Precision.** Type-1 NUFFT over the cloud with weights $\Delta_i/(S\sigma_r^2)$ to get the BTTB generator T_r; A_r acts as $v \mapsto v + D_\theta T_r D_\theta v$.
4. **RHS.** Type-1 NUFFTs over the cloud with weights $d_{i,r}/(S\sigma_r^2)$ and $[C_i]_{j,r}/(S\sigma_r^2)$ produce $h_{1,r}$ and $h_{2,r}$ via (33).
5. **Solve.** Jacobi-preconditioned CG: $A_r \mu_r = h_r$.
6. **Predict.** For each evaluation point (typically $x = m_i$), one Type-2 NUFFT each gives $\bar{f}(x)$ and $J_{\bar{f}}(x)$; Hutchinson probes give $\operatorname{diag}(\Phi A_r^{-1} \Phi^*)$.
7. $q(x)$ **update.** Plug $\bar{f}$, $J_{\bar{f}}$, and the second moment into SING's natural-gradient update via `FrozenEFGPDrift`'s custom VJPs.

10 Complexity

Let $N = (T-1)S$ be the cloud size, M the number of Fourier features, k the number of CG iterations, and J the number of Hutchinson probes. The $q(f)$ update costs

$$\underbrace{O(N)}_{\text{cloud assembly}} \quad + \quad \underbrace{O(M \log M)}_{\text{NUFFTs / BTTB}} \quad + \underbrace{O(k\, D_x\, M \log M)}_{\text{CG over output dims}}. \tag{45}$$

Drift moments at T points cost $O(T) + O(D_x M \log M)$; Hutchinson variance adds $O(J\, k\, M \log M)$. The implementation rule is to never materialise K, Φ, $\Phi^* W \Phi$, A_r^{-1}, or any dense covariance matrix: every required action is a Fourier matvec, a BTTB matvec, a CG solve, or a Hutchinson probe.

A Math-to-code dictionary

Math	Code
Spectral grid weights $D_\theta = \mathrm{diag}(\sqrt{S(\xi_k)\, h^d})$	`grid.ws` (real part)
Frequency points ξ_k	`grid.xis_flat`
Adjoint Type-1 NUFFT F_X^*	`nufft1`
Forward Type-2 NUFFT $\phi(x)^*\mu$	`nufft2`
BTTB generator $T_r = F_X^* W_r F_X$	`bttb_conv_vec_weighted` $\rightarrow$ `make_toeplitz`
Operator $A_r v = v + D_\theta T_r D_\theta v$	`make_A_apply`
Solve $A_r \mu_r = h_r$ (Jacobi-preconditioned CG)	`cg_solve`
$h_r = h_{1,r} + h_{2,r}$ build	`compute_mu_r_jax`
$\bar{f}(x)$, $J_{\bar{f}}(x)$, $\mathbb{E}[f^\top \Sigma^{-1} f]$	`drift_moments_jax`
$\mathrm{diag}(\Phi A_r^{-1} \Phi^*)$ (Hutchinson)	`hutchinson_diag`
Local-quadratic VJPs into $q(x)$ update	`ef_with_jac_grad`, `eff_with_grads`
$q(x)$ consumer	`nat_grad_transition` (`sing/sing.py`)

$q(x)$ statistic	Code
m_i (marginal mean)	`ms`
S_i (marginal covariance)	`Ss`
$\mathrm{Cov}(x_i, x_{i+1}) = S_{i,i+1}$	`SSs.transpose(0,2,1)`
$d_i = m_{i+1} - m_i$	`d_a`
$C_i = S_{i,i+1} - S_i$	`C_a`